Beyond The SIX SENSES

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Innovations such as drones, robots and satellites have significantly inproved the level of security of defence armed forces. In this article we learn about the sensors within and beyond those used in the military sector

Sensors have radically changed how we understand and interact with our world in real time. These allow us to detect data and other key performance metrics that can be analysed to get accurate information for weather forecast, automation, defence, military and aerospace.

Fig. 1: First-day image from Cartosat-2 series satellite (Credit: www.isro.gov.in)

A sensor senses changes in the environment and sends information to other electronics. Its sensitivity indicates how much its output changes when the input quantity being measured changes. In this article we talk about sensor applications in defence using remote sensing, weather monitoring, robots and drones.

Central Reserve Police Force (CRPF) has started procuring drones for tackling Maoists and terrorists. Defense Research and Development Organisation (DRDO) successfully tested Rustom 2, a heavy-duty drone. The objective of this drone is to carry out 24-hour surveillance for the armed forces.

Cartosat-2 series - PSLV-C40 satellite was launched from SDSC SHAR, Sriharikota, Andhra Pradesh, on January 12, 2018 at 09:29 hours (IST). It is a follow-on mission of Cartosat-2 series with the primary objective of providing high-resolution scene-specific spot imageries. It was made by Indian Space Research Organisation (ISRO) for observating the Earth revolving in Sun's synchronous polar orbit. The satellite carries panchromatic (black and white) and multi-spectral (colour) cameras, thus capable of delivering high-resolution data.

Remote sensing

Remote sensing can be active or passive based on the sensors used. Active sensors send light waves from their own source and measure the backscatter reflected light. On the other hand, passive sensors measure reflected sunlight (emitted from the Sun) from the objects.

Remote sensing helps defence, military and aerospace to provide different kinds of information. Navigating ships use remote sensing technologies like wind wave information, routing analysis, ship proximity along with GPS to save ships from sinking on hitting an iceberg. For searching crashed aircraft, many satellites orbit the Earth each day, collecting